

1. General Description

The EMS3157 is a single SPDT low on-resistance analog switch. It can operate from 1.5V to 5.5V power supply. The device offers low ON-state resistance and excellent ON-state resistance matching with break-before-make feature, to prevent signal distortion during the transferring of a signal from one channel to another. The device is capable of truly isolation. Even when A overrides V_{CC} , very little current will flow back to the supply.

2. Features and Benefits

- Wide Supply Range from 1.5 V to 5.5 V
- Low ON-state resistance, typically $1.9\ \Omega$ ($V_{CC}=5.5\text{ V}$)
- 1.8 V Compatible Control Logic Inputs
- High Off-Isolation: -80 dB @ 10 kHz
- Low Channel-to-Channel Crosstalk: -82 dB @ 10 kHz
- High Bandwidth: -3dB @ 800MHz
- Low Quiescent Current ($< 1\mu\text{A}$) when $V_{CC} = 3.3\text{ V}$
- Latch-up performance exceeds 200 mA
- ESD protection:
 - HBM ANSI/ESDA/JEDEC JS-001 Class 2 exceeds 3000 V
 - CDM ANSI/ESDA/JEDEC JS-002 Class C3 exceeds 2000 V

3. Applications

- Audio, Video, UART, USB2.0 Signal and Supply Routing
- Cell phones and TWS headset

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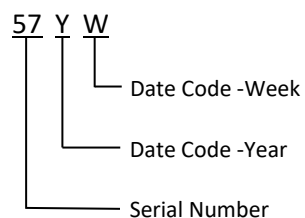
4. Ordering Information

Table 1. Ordering information

Type number	Topside marking	Package		
		Name	Description	Quantity
EMS3157GW	57 YW	SOT363	SOT363 package, 6 pins 2.1 mm × 1.25 mm; 1.1 mm (Max) height	3000
EMS3157GV	57 YW	SOT23-6L	SOT23 package, 6pins 2.92 mm × 1.6 mm; 1.25 mm (Max) height	3000
EMS3157GM	57 YW	DFN1.45x1-6L	DFN1.45×1 package, 6 pins 1.45 mm × 1 mm; 0.6 mm (Max) height	3000
EMS3157GS	57	DFN1x1-6L	DFN1×1 package, 6 pins 1 mm × 1 mm; 0.42 mm (Max) height	3000
EMS3157GX3	57	DFN0.8X1-6L	DFN0.8×1 package, 6 pins 0.8 mm × 1 mm; 0.35 mm (Max) height	3000

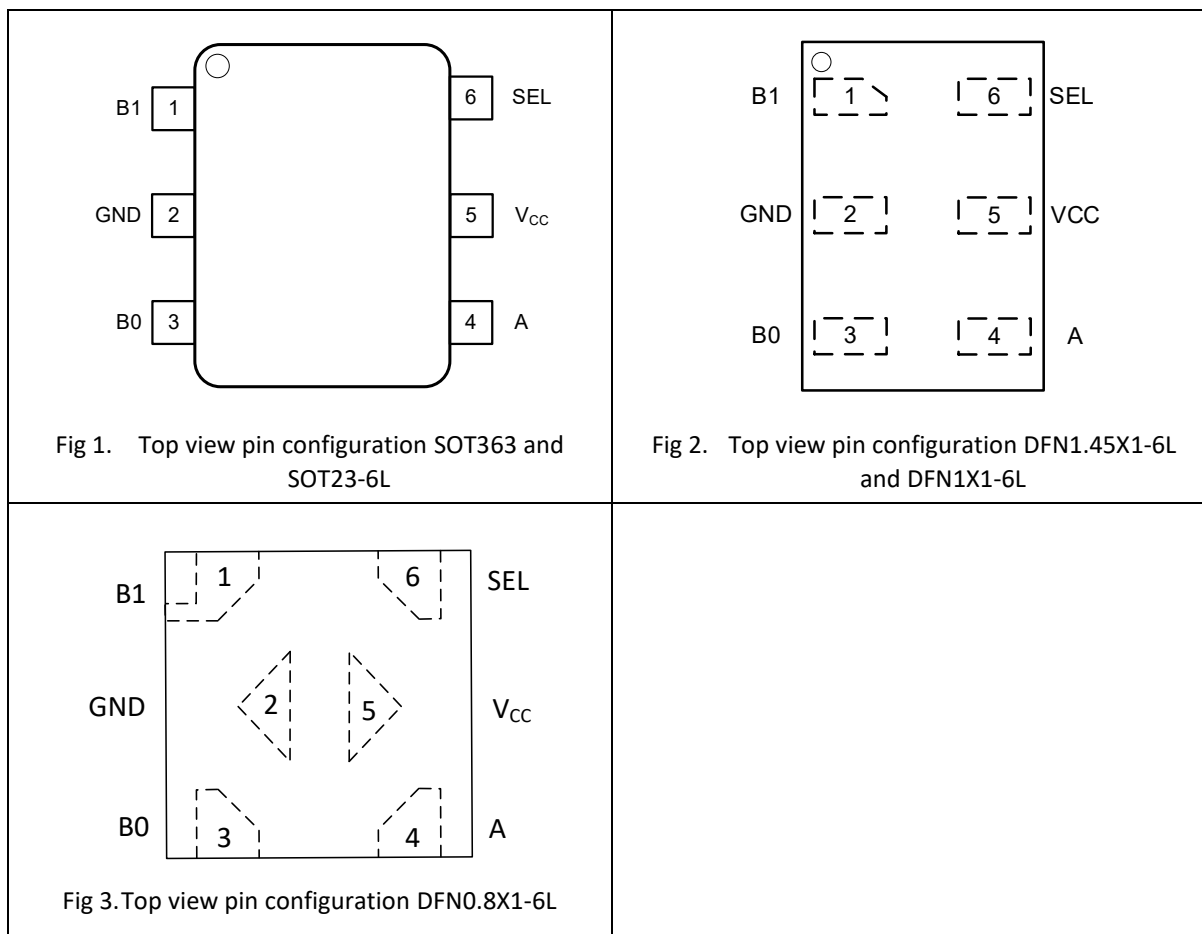
MARKING INFORMATION

NOTE: YW = Date Code.



5. Pinning Information

5.1. Pin map



5.2. Pin description

Table 2. Pin description

Symbol	Pin	Description
B1	1	Analog/Digital Signal Port (Normally open)
GND	2	Ground (0V)
B0	3	Analog/Digital Signal Port (Normally closed)
A	4	Common Signal Port
V _{CC}	5	Supply voltage
SEL	6	Logic Input Control

6. Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level.

Input SEL	Channel on
L	B0
H	B1

7. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

Table 4. Absolute Maximum Ratings

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

Symbol	Parameter	Conditions	Min	Max	Unit
V _{CC}	supply voltage		-0.3	6.5	V
V _{SEL}	control Input Voltage		-0.3	6.5	V
V _A	signal input voltage		-0.3	6.5	V
I _{max}	continuous Current Through A, B0, B1			±100	mA
	peak Current Through A, B0, B1 (pulsed at 1ms 50% duty cycle)			±200	mA
T _J	junction Temperature under Bias			150	°C
T _L	lead Temperature (Soldering, 10 seconds)			260	°C
R _{θJA}	thermal resistance			350	°C/W

8. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

Table 5. Recommended Operating Conditions

Symbol	Parameter	Conditions	Min	Max	Unit
V _{CC}	supply voltage		1.5	5.5	V
V _{SEL}	control input voltage		0	5.5	V
V _A	signal input voltage		0	V _{CC}	V
T _{amb}	ambient temperature		-40	125	°C

9. Electrical Characteristics

Table 6. Static characteristics
 $T_{amb}=25\text{ }^{\circ}\text{C}$, $V_{CC}=3.3\text{ V}$, unless otherwise specified

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
DC Electrical Characteristics						
V_{IH}	input logic high level	$V_{CC} = 3.4\text{ V to }5.5\text{ V}$	1.6			V
		$V_{CC} = 1.5\text{ V to }3.3\text{ V}$	1.4			V
V_{IL}	input logic low level	$V_{CC} = 3.4\text{ V to }5.5\text{ V}$			0.6	V
		$V_{CC} = 1.5\text{ V to }3.3\text{ V}$			0.4	V
I_{CC}	supply current	$I_A = 0$, $V_{SEL} = 0$ or $V_{SEL} = V_{CC}$			1.0	μA
ΔI_{CC}	additional supply current	$I_A = 0$, $V_{CC} = 2.5\text{ V or }4.2\text{ V}$ $V_{SEL} = 1.6\text{ V}$			3.0	μA
I_{off}	OFF-state switch leakage current from A to B0 (or B1)	$V_A = 5\text{ V}$, V_{B0} or $V_{B1} = 0\text{ V}$			± 2.0	μA
R_{ON}	on-resistance	$V_{CC} = 3.5\text{ V}$, $V_A = 0.8\text{ V to }2.0\text{ V}$, $I_A = 10\text{ mA}$		3.4	4.0	Ω
		$V_{CC} = 4.5\text{ V}$, $V_A = 0.8\text{ V to }2.3\text{ V}$, $I_A = 10\text{ mA}$		2.4	3.0	Ω
		$V_{CC} = 5.5\text{ V}$, $V_A = 0.8\text{ V to }2.6\text{ V}$, $I_A = 10\text{ mA}$		1.9	2.5	Ω
R_{FLAT}	on-resistance flatness	$V_{CC} = 3.5\text{ V}$, $V_A = 0.8\text{ V to }2.0\text{ V}$, $I_A = 10\text{ mA}$		0.2	0.5	Ω
		$V_{CC} = 4.5\text{ V}$, $V_A = 0.8\text{ V to }2.3\text{ V}$, $I_A = 10\text{ mA}$		0.25	0.5	Ω
		$V_{CC} = 5.5\text{ V}$, $V_A = 0.8\text{ V to }2.6\text{ V}$, $I_A = 10\text{ mA}$		0.3	0.5	Ω
ΔR_{ON}	on-resistance matching between channels	$V_{CC} = 3.5\text{ V}$, $V_A = 0.8\text{ V to }2.0\text{ V}$, $I_A = 10\text{ mA}$		0.1	0.2	Ω
		$V_{CC} = 4.5\text{ V}$, $V_A = 0.8\text{ V to }2.3\text{ V}$, $I_A = 10\text{ mA}$		0.1	0.2	Ω
		$V_{CC} = 5.5\text{ V}$, $V_A = 0.8\text{ V to }2.6\text{ V}$, $I_A = 10\text{ mA}$		0.1	0.2	Ω

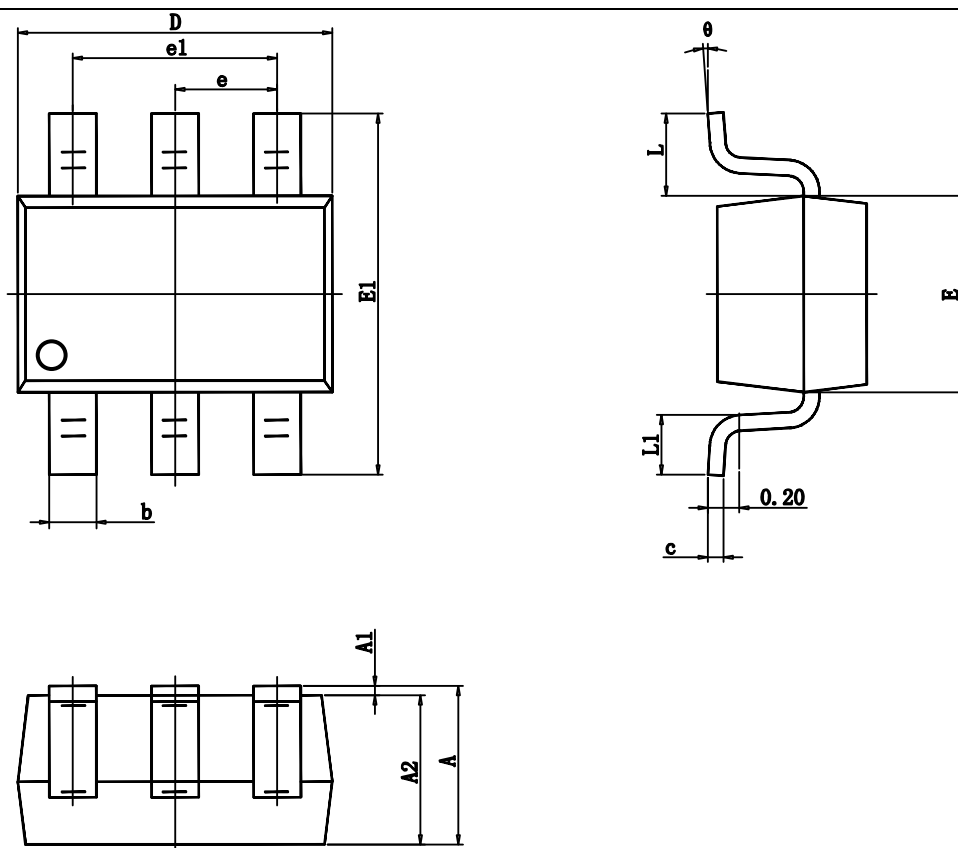
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Symbol	Parameter	Conditions	Min	Typ	Max	Unit
AC Electronics Characteristics						
t_{ON}	turn-On Time	$V_A=1.5V, C_L=50pF, R_L=50\Omega$		470		ns
t_{OFF}	turn-Off Time	$V_A=1.5V, C_L=50pF, R_L=50\Omega$		120		ns
t_{BBM}	break-before-make time	$V_A=1.5V, C_L=50pF, R_L=50\Omega$		600		ns
BW	-3dB Bandwidth	$C_L=0pF$		800		MHz
OIRR	off isolation	$f=10\text{ KHz}$		-80		dB
		$f=240\text{ MHz}$		-31		dB
X_{TALK}	crosstalk	$f=10\text{ KHz}$		-82		dB
		$f=240\text{ MHz}$		-26		dB
THD	total harmonic distortion	$f=20\text{Hz to } 20\text{KHz}$ $V_A=600mV_{pp}, R_L=32\Omega$		-80		dB
Capacitance						
C_{ON}	On Capacitance	$f=100\text{KHz}$		12		pF
C_{OFF}	Off Capacitance	$f=100\text{KHz}$		4		pF

10. Package Outline

SOT363

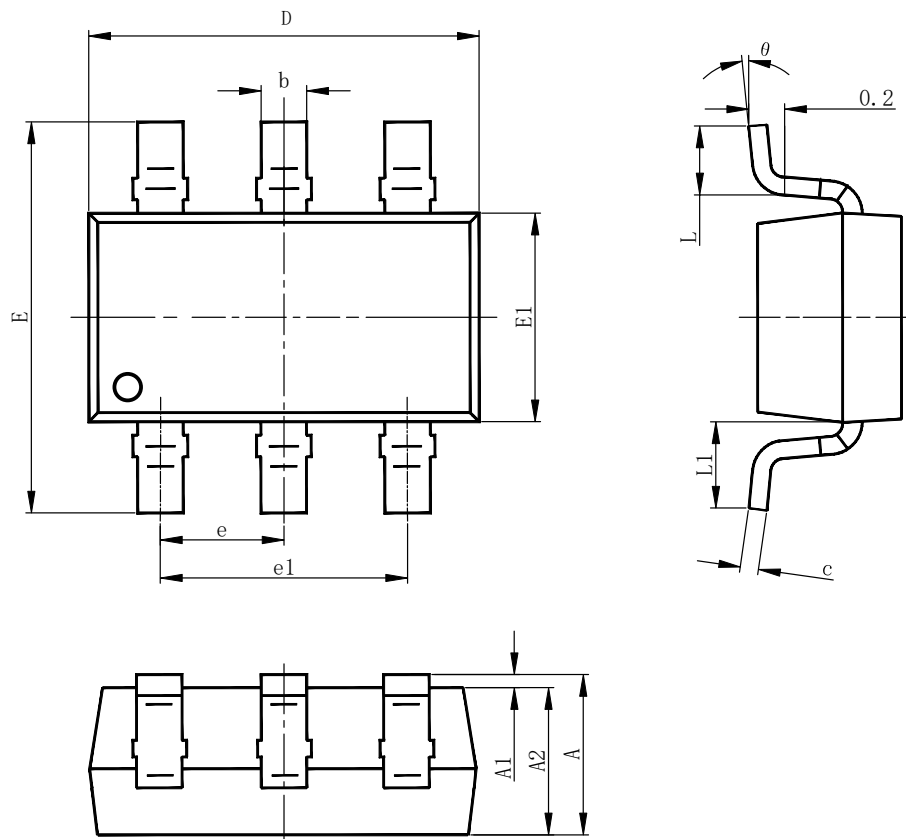


Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	0.900	1.100	0.035	0.043
A1	0.000	0.100	0.000	0.004
A2	0.900	1.000	0.035	0.039
b	0.150	0.350	0.006	0.014
c	0.110	0.175	0.004	0.007
D	2.000	2.200	0.079	0.087
E	1.150	1.350	0.045	0.053
E1	2.150	2.450	0.085	0.096
e	0.650 TYP.		0.026 TYP.	
e1	1.200	1.400	0.047	0.055
L	0.525 REF.		0.021 REF.	
L1	0.260	0.460	0.010	0.018
θ	0°	8°	0°	8°

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SOT23-6L

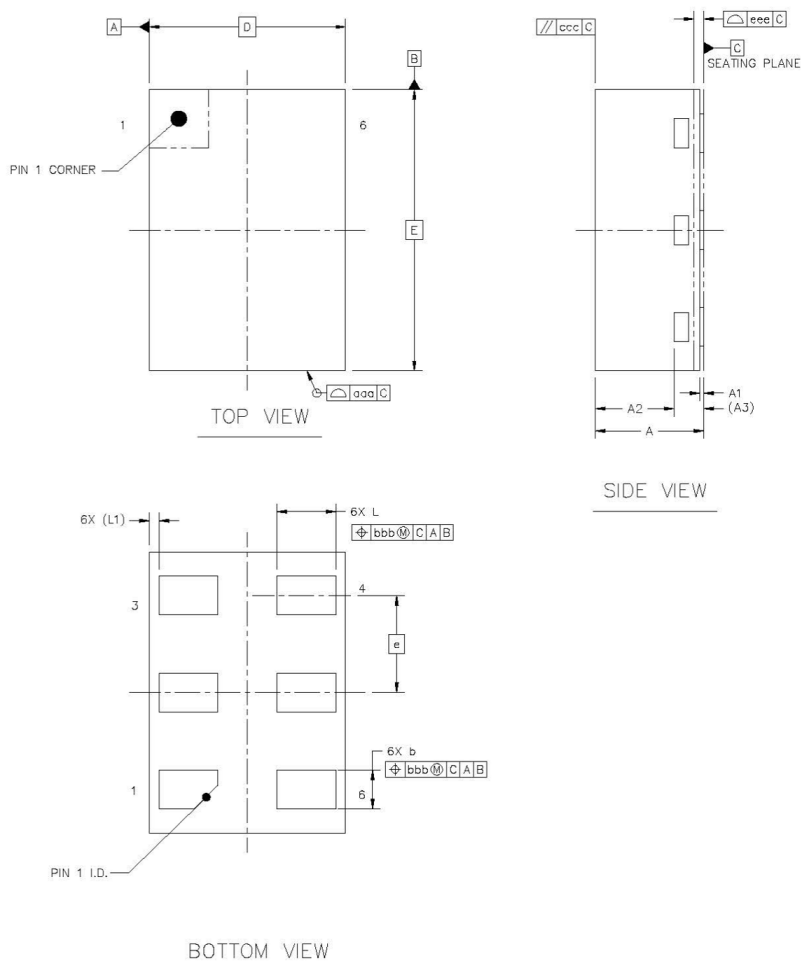


Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	1.050	1.250	0.041	0.049
A1	0.000	0.100	0.000	0.004
A2	1.050	1.150	0.041	0.045
b	0.300	0.500	0.012	0.020
c	0.100	0.200	0.004	0.008
D	2.820	3.020	0.111	0.119
E1	1.500	1.700	0.059	0.067
E	2.650	2.950	0.104	0.116
e	0.950(BSC)		0.037(BSC)	
e1	1.800	2.000	0.071	0.079
L	0.300	0.600	0.012	0.024
L1	0.600REF.		0.024REF.	
θ	0°	8°	0°	8°

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DFN1x1.45-6L

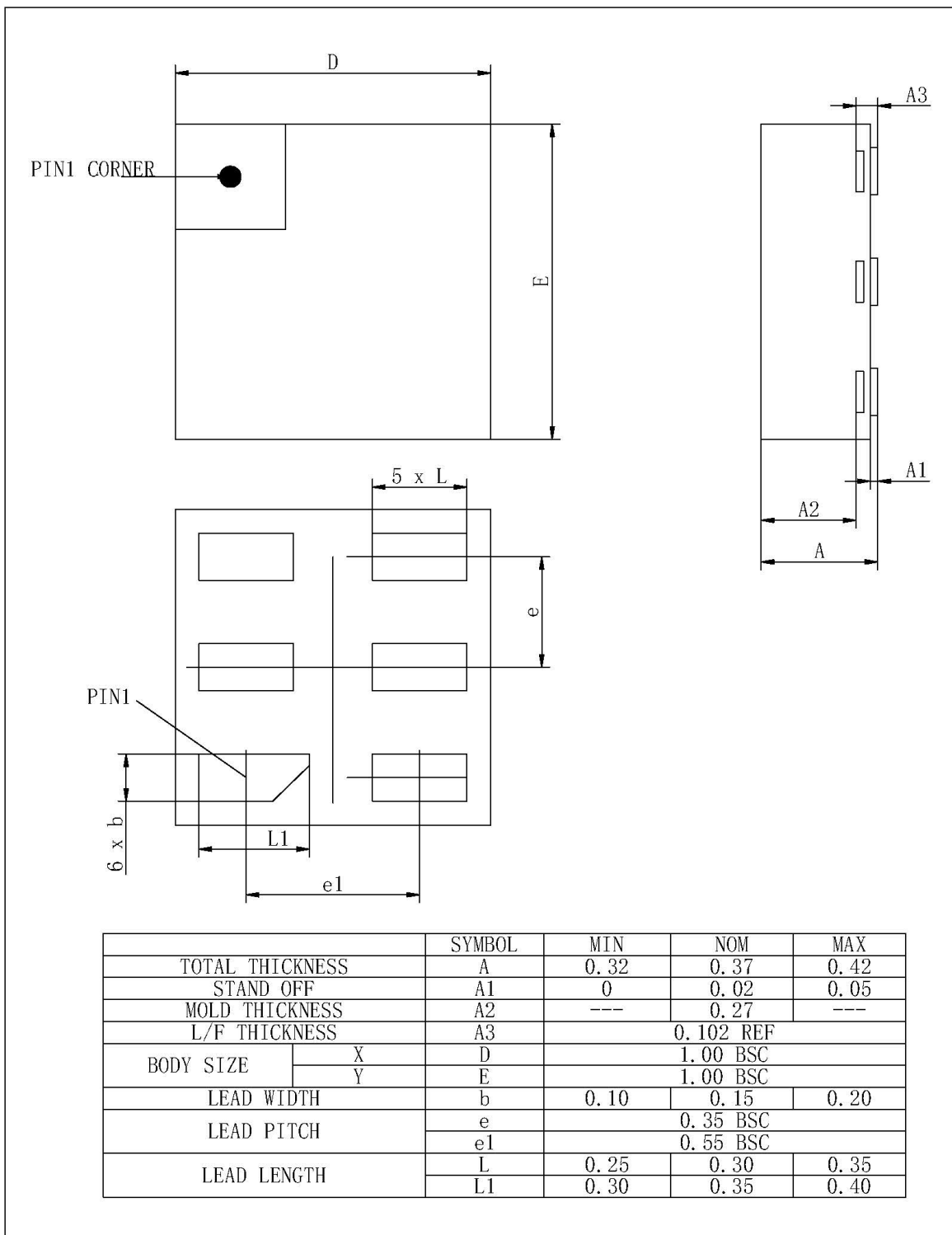


		SYMBOL	MIN	NOM	MAX
TOTAL THICKNESS		A	0.5	0.55	0.6
STAND OFF		A1	0	0.02	0.05
MOLD THICKNESS		A2	---	0.4	---
L/F THICKNESS		A3	0.152 REF		
LEAD WIDTH		b	0.15	0.2	0.25
BODY SIZE	X	D	1 BSC		
	Y	E	1.45 BSC		
LEAD PITCH		e	0.5 BSC		
LEAD LENGTH		L	0.25	0.3	0.35
		L1	0.05 REF		
PACKAGE EDGE TOLERANCE		aaa	0.1		
MOLD FLATNESS		ccc	0.1		
COPLANARITY		eee	0.05		
LEAD OFFSET		bbb	0.1		

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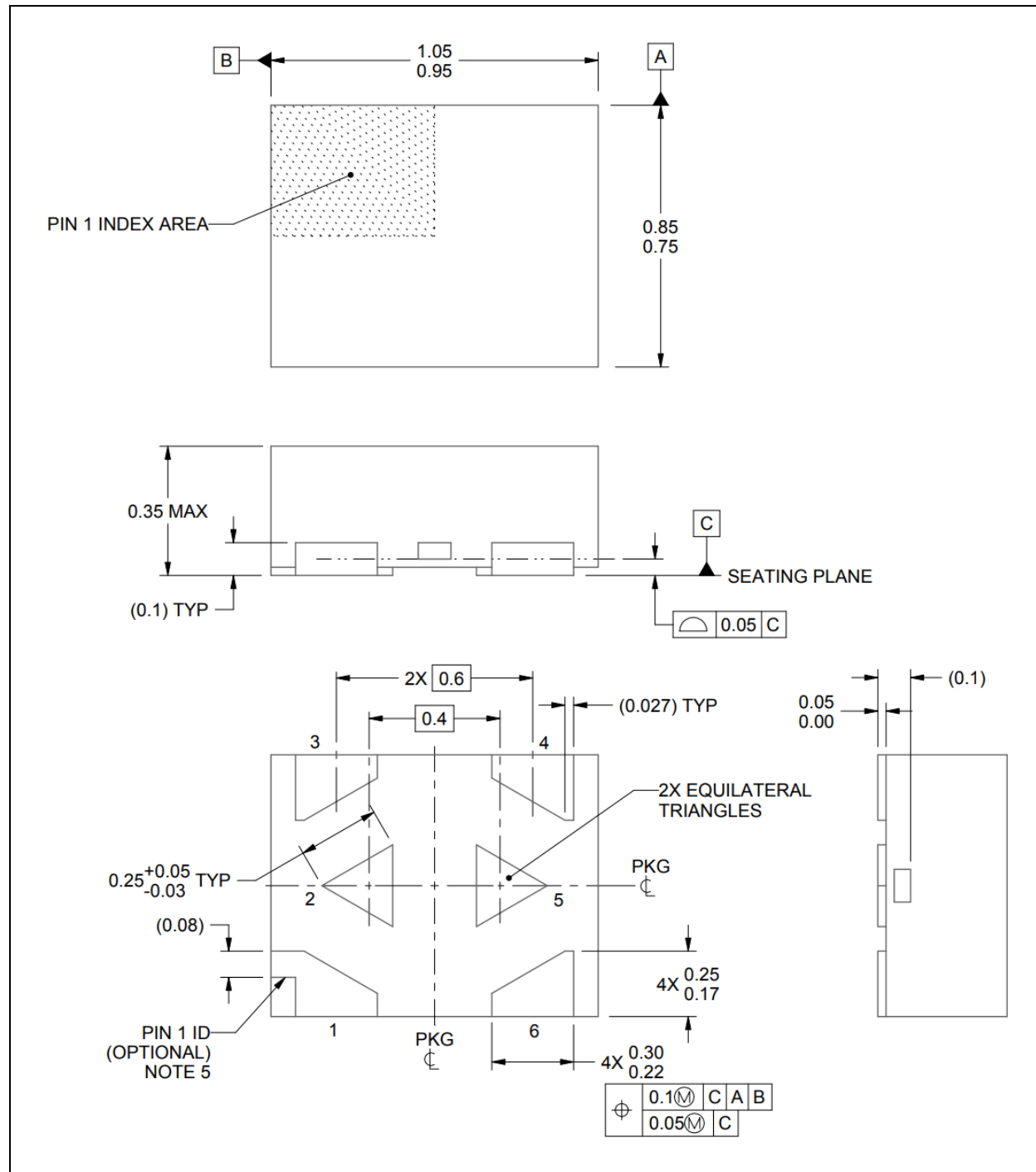
DFN1x1-6L



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DFN0.8x1-6L



11. Tape and Reel Information

11.1. Carrier tape dimensions

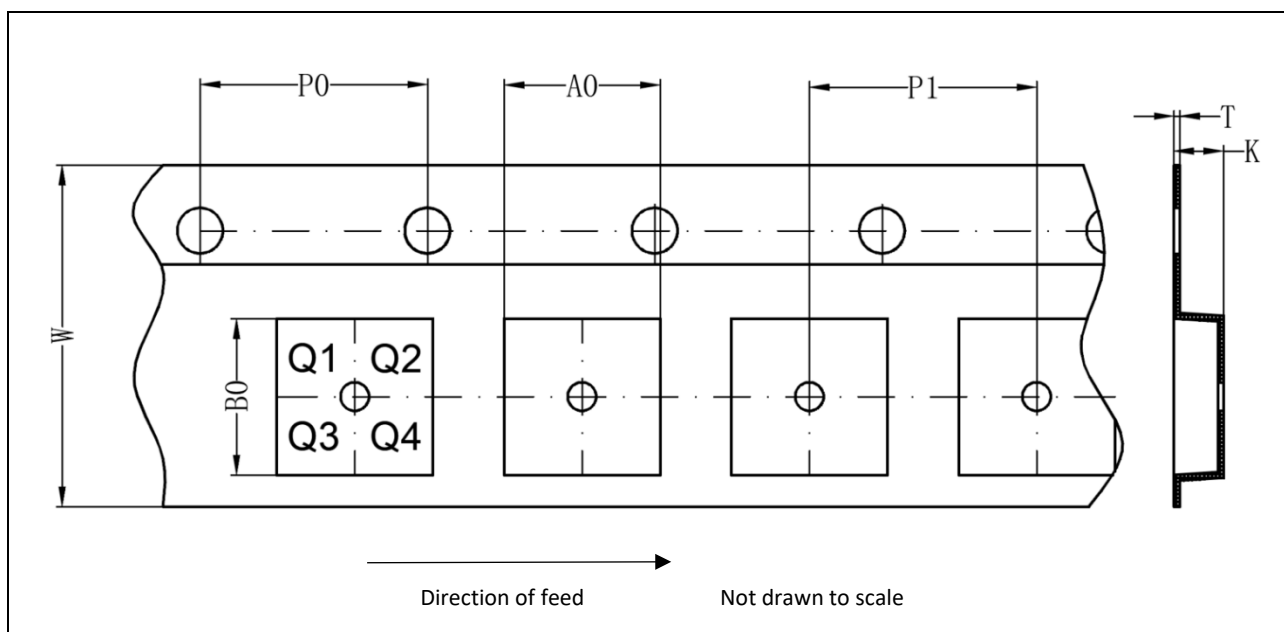
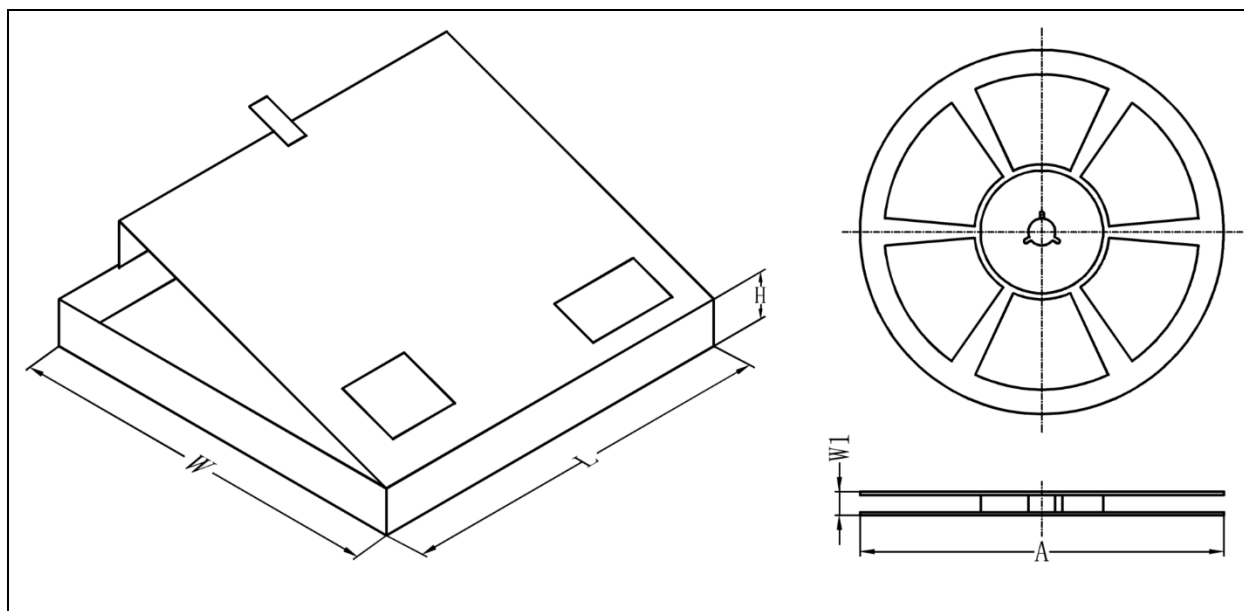


Table 7. Carrier tape dimensions

Package version	A0(mm)	B0(mm)	K(mm)	T(mm)	P1(mm)	W(mm)	P0(mm)	PIN 1
SOT363	2.55	2.55	1.2	0.2	4	8	4	Q3
SOT23-6L	3.23	3.17	1.37	0.25	4	8	4	Q3
DFN1x1.45-6L	1.15	1.6	0.75	0.2	4	8	4	Q1
DFN1x1-6L	1.16	1.16	0.5	0.23	4	8	4	Q1

11.2. Reel and box dimensions


Table 8. Dimensions and quantities

Package version	Type NO. ending	Reel Dimension A (mm)	Max Reel Width W1 (mm)	SPQ (pcs)[1]	Reels per box	Outer box dimensions L×W×H(mm)[2]
SOT363	GW	180	13.2	3000	1	210x200x40
SOT23-6L	GV	180	13.2	3000	1	210x200x40
DFN1x1.45-6L	GM	180	13.2	3000	1	210x200x40
DFN1x1-6L	GS	180	13.2	3000	1	210x200x40

[1] Packing quantity dependent on specific product type. Please contact your local Energymath representative for ordering.

[2] Dimensions for reference only.

12. Revision History

Table 9. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EMS3157 Rev. 1.8	Jul 10, 2026	Product datasheet		EMS3157 Rev. 1.7
Modifications:	- Added DFN0.8x1-6L package. - Added ESD protection information.			
EMS3157 Rev. 1.7	May 15, 2026	Product datasheet		EMS3157 Rev. 1.6
Modifications:	Add DFN1x1-6L package.			
EMS3157 Rev. 1.6	Aug 12, 2025	Product datasheet		EMS3157 Rev. 1.5
Modifications:	- Section 4: Add marking information. - Section 5.1: Fig 1 updated. - Table 6 : R_{FLAT} updated. - Add DFN1.45x1 package. - Section 11: Add tape and reel information.			
EMS3157 Rev. 1.5	Mar 20, 2025	Product datasheet		EMS3157 Rev. 1.4
Modifications:	Table 6 : OIRR and X_{TALK} updated.			
EMS3157 Rev. 1.4	Feb 13, 2025	Product datasheet		EMS3157 Rev. 1.3
Modifications:	- Features and Benefits updated. - Table 6: On resistance updated.			
EMS3157 Rev. 1.3	Dec 13, 2024	Product datasheet		EMS3157 Rev. 1.2
Modifications:	Table 1: Topside marking updated.			
EMS3157 Rev. 1.2	Dec 10, 2024	Product datasheet		EMS3157 Rev. 1.1
Modifications:	Table 6: ΔI_{CC} and AC Characteristics updated.			
EMS3157 Rev. 1.1	Nov 30, 2024	Product datasheet		EMS3157 Rev. 1.0
Modifications:	Add SOT23-6L package.			
EMS3157 Rev. 1.0	Oct 30, 2023	Product datasheet		